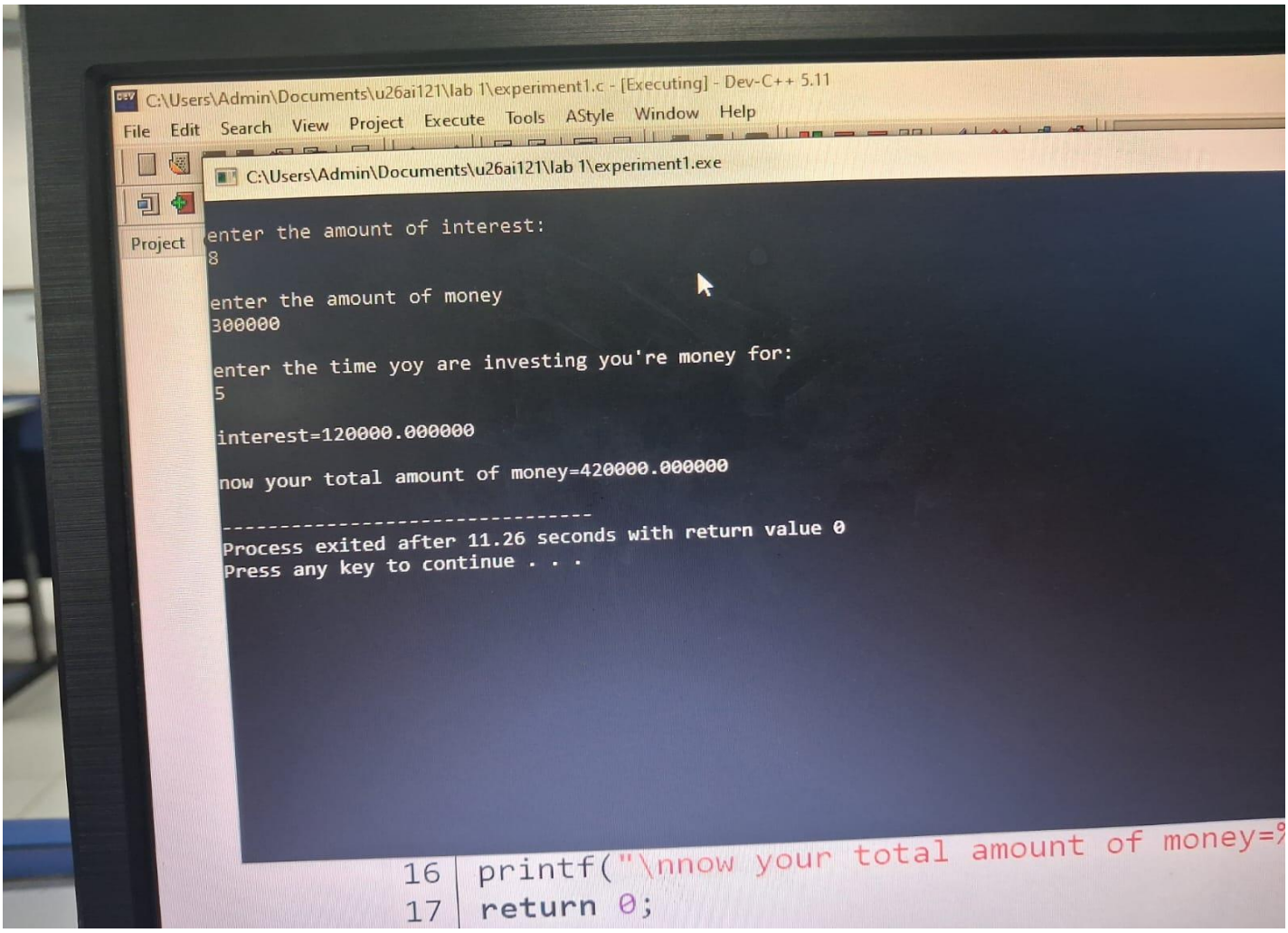
Q.1

```c
// Online C compiler to run C program online
#include <stdio.h>

int main() {
    // Write C code here
    float money,interest_rate,time;
    printf("\nenter the amount of interest:\n");
    scanf("%f",&interest_rate);
    printf("\nenter the amount of money\n");
scanf("%f",&money);
printf("\nenter the time yoy are investing you're money for:\n");
scanf("%f",&time);
float interest=(money*interest_rate*time)/100;
printf("\ninterest=%f\n",interest);
float total=money+interest;
printf("\nnow your total amount of money=%f\n",total);
return 0;
}

```



Q.2

```c
#include <stdio.h>

int main() {
    // Write C code here
float sub1,sub2,sub3,sub4,sub5;
printf("enter marks of maths=");
scanf("%f",&sub1);
printf("enter marks of english=");
scanf("%f",&sub2);

printf("enter marks of science=");
scanf("%f",&sub3);

printf("enter marks of sst=");
scanf("%f",&sub4);
printf("enter marks of sanskrit=");
scanf("%f",&sub5);
float percentage=(sub1+sub2+sub3+sub4+sub5)/5;
printf("your percentage is:%f",percentage);
    return 0;
}
```

```
C:\Users\Admin\Documents\u26ai121\lab 1\experiment1.2.exe
enter marks of maths=50
enter marks of english=56
enter marks of science=67
enter marks of sst=89
enter marks of sanskrit=98
your percentage is:72.000000
--------------------------------
Process exited after 10.21 seconds with return value 0
Press any key to continue . . .
```

Q.3

10:38

Compile Result

```
enter n150000
your gross salary is 123000.000000
[Process completed - press Enter]
```

10:38

Coding C

Nothing changed

RUN MENU

```c
#include <stdio.h>
#include<math.h>
void gross(float n);
int main() {
float n;
printf("enter n");
scanf("%f",&n);
gross(n);
return 0;}
void gross(float n){

printf("your gross salary is %f",(0.82)*n);
}
```

Q.4

Coding C

Nothing changed

RUN MENU

```c
(sum==t)(printf("it is an armstrong number");)
se(printf("it is not armstrong number");)



turn 0;)
 */
 #include <stdio.h>

t main() {
 float fahrenheit, centigrade;


 printf("Enter temperature in Fahrenheit: ");
 scanf("%f", &fahrenheit);


 centigrade = (fahrenheit - 32) * 5 / 9;


 printf("%.2f Fahrenheit is equal to %.2f Centigrade.\n", fahrenheit, centigrade)

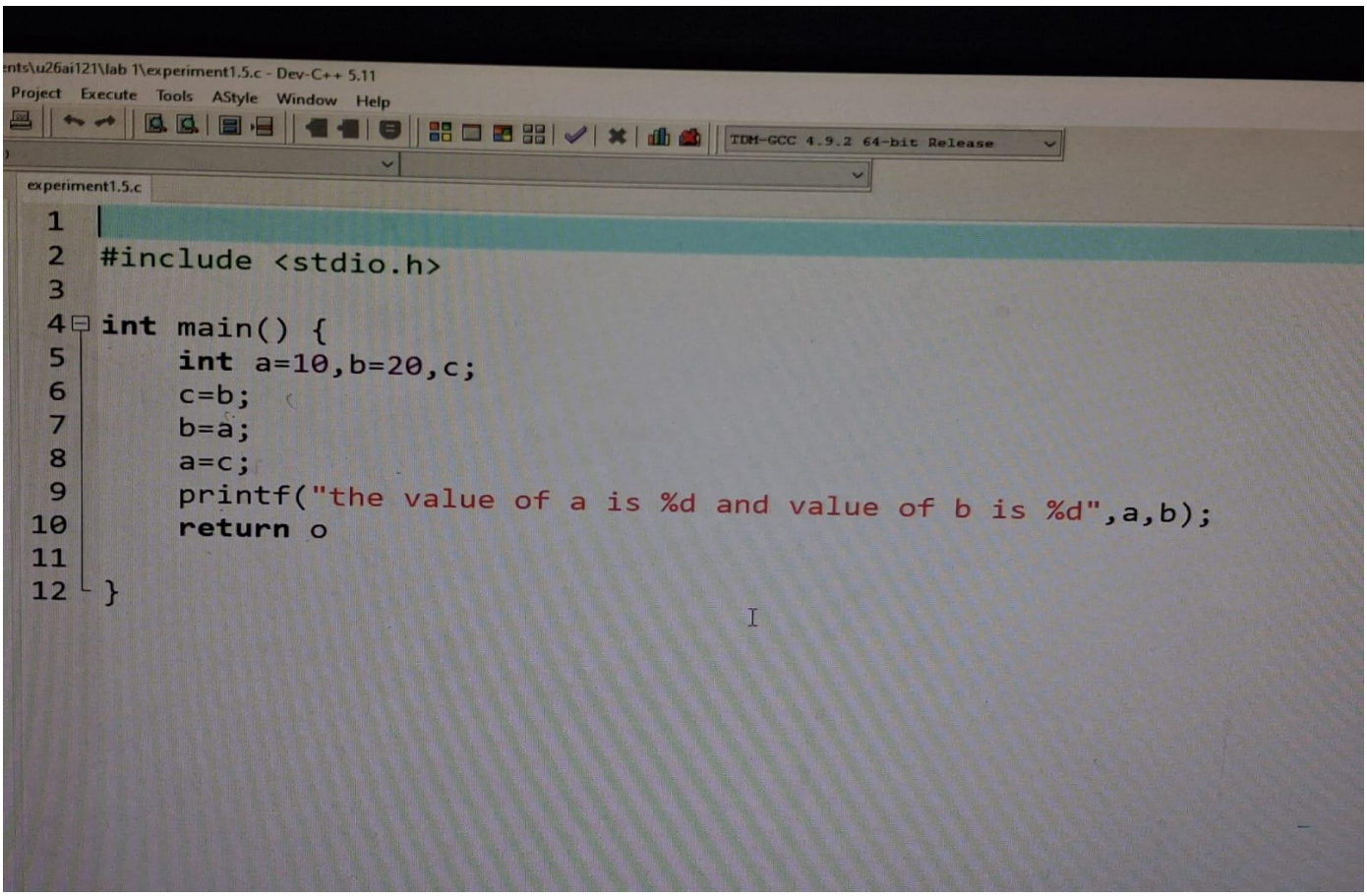
 return 0;
```

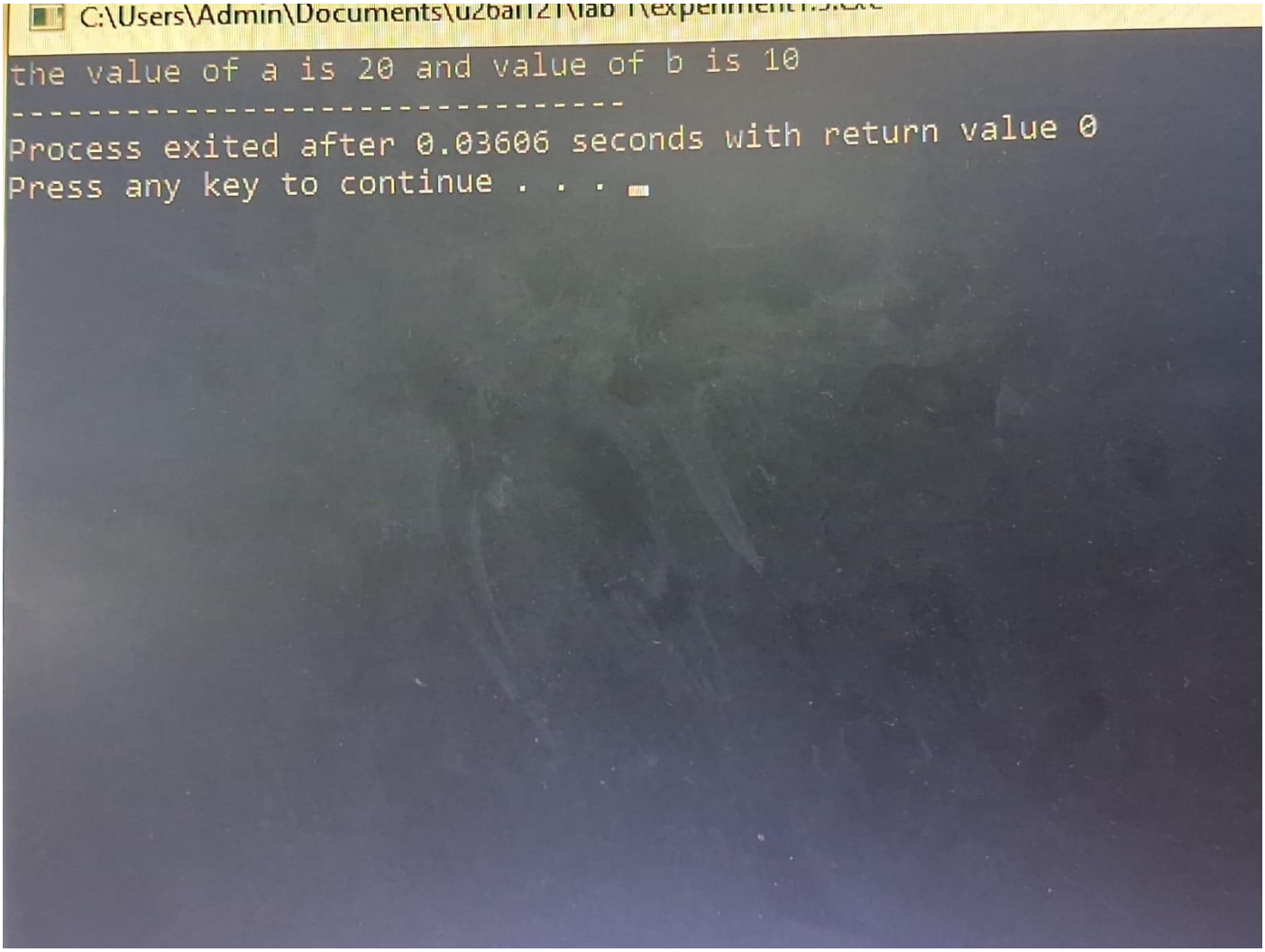
Compile Result

```
Enter temperature in Fahrenheit: 67
67.00 Fahrenheit is equal to 19.44 Centigrad
e.

[Process completed - press Enter]
```

Q.5


nts\u26ai121\lab 1\experiment1.5.c - Dev-C++ 5.11
Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
experiment1.5.c
1
2 #include <stdio.h>
3
4 int main() {
5 int a=10,b=20,c;
6 c=b;
7 b=a;
8 a=c;
9 printf("the value of a is %d and value of b is %d",a,b);
10 return o
11
12 }


C:\Users\Admin\Documents\u26ai121\lab 1\experiment1.5.exe
the value of a is 20 and value of b is 10

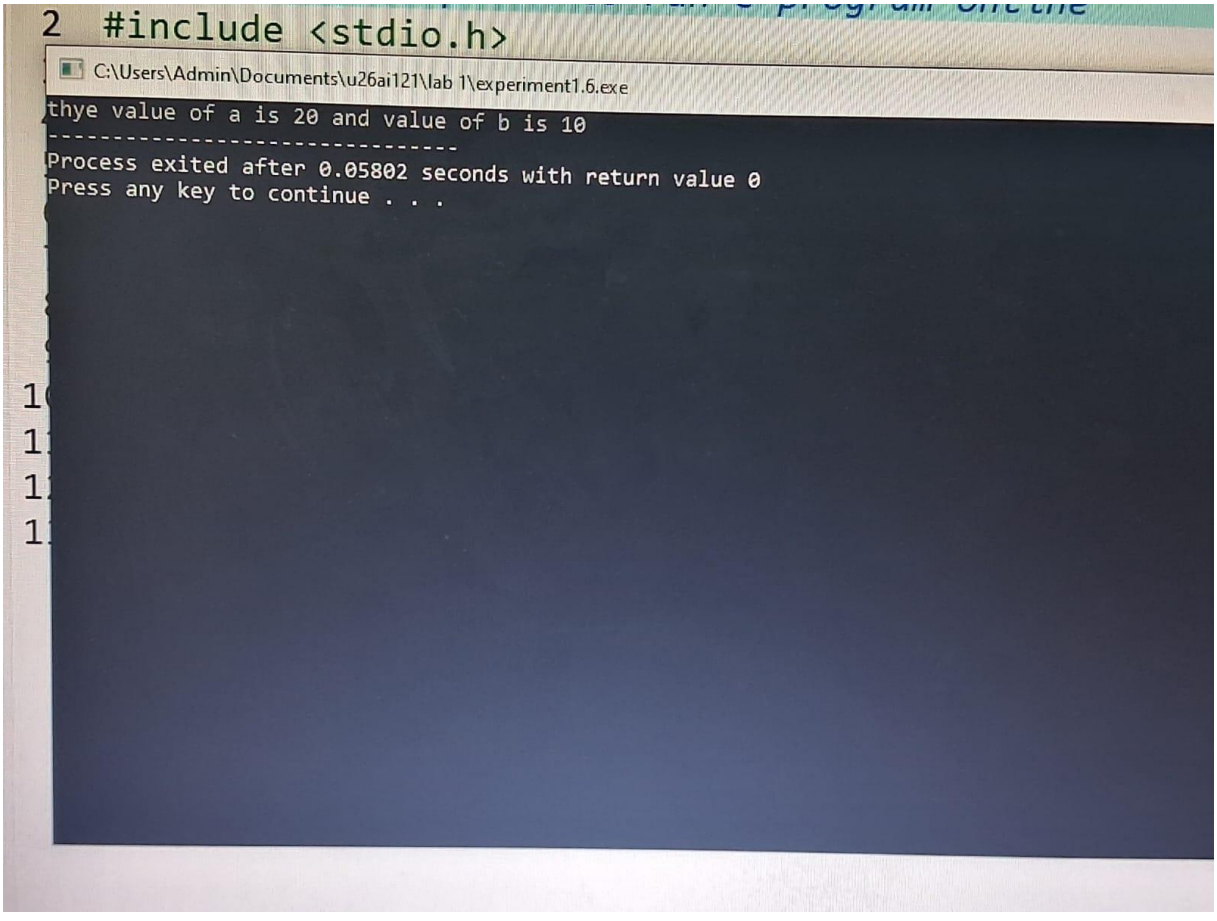
Process exited after 0.03606 seconds with return value 0
Press any key to continue . . .

Q.6

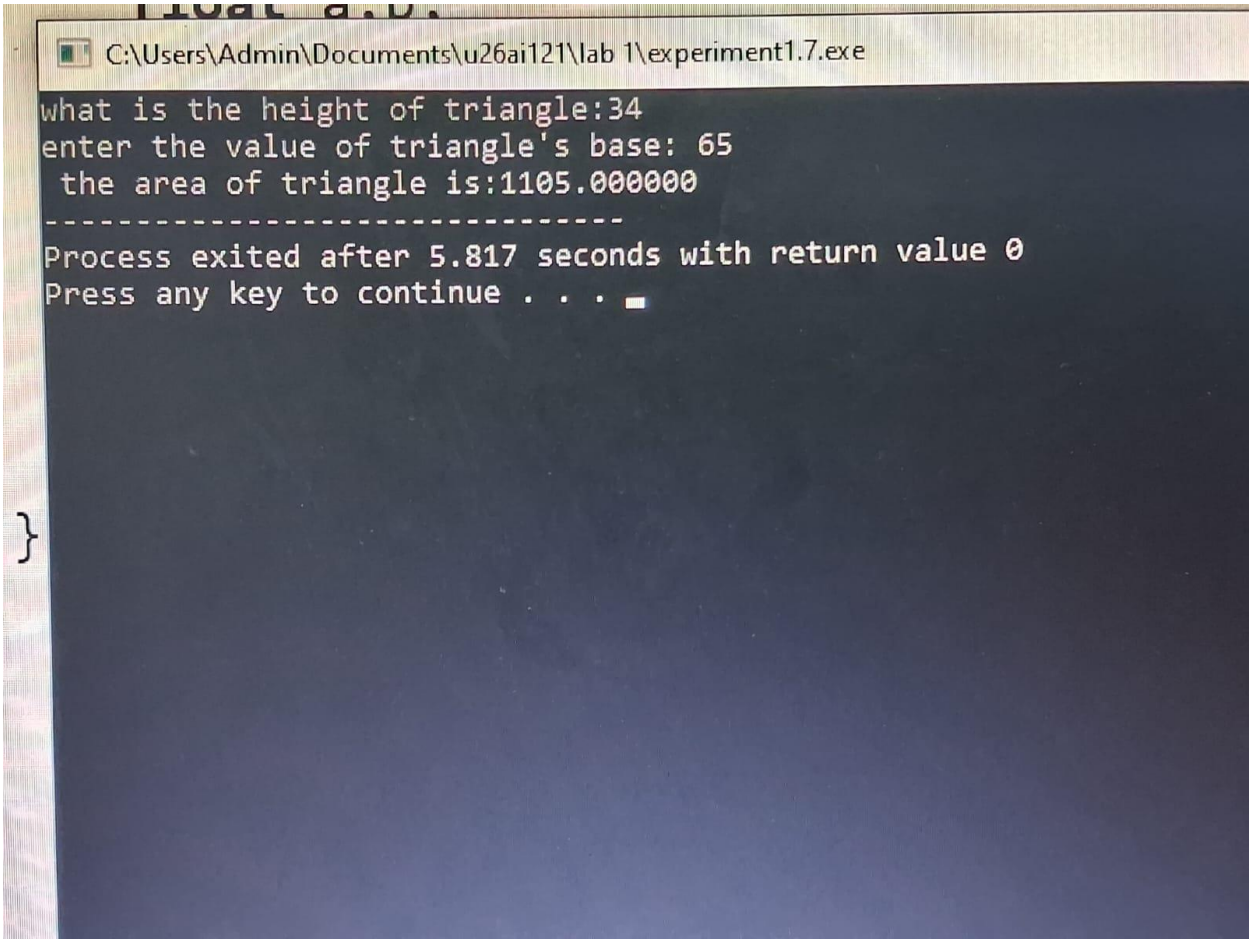
```c
// Online C compiler to run C program online
#include <stdio.h>

int main() {
    int a=10,b=20;
    a=a^b;
        b=a^b;
        a=a^b;
    // Write C code here
    printf("thye value of a is %d and value of b is %d",a,b);

    return 0;
}
```

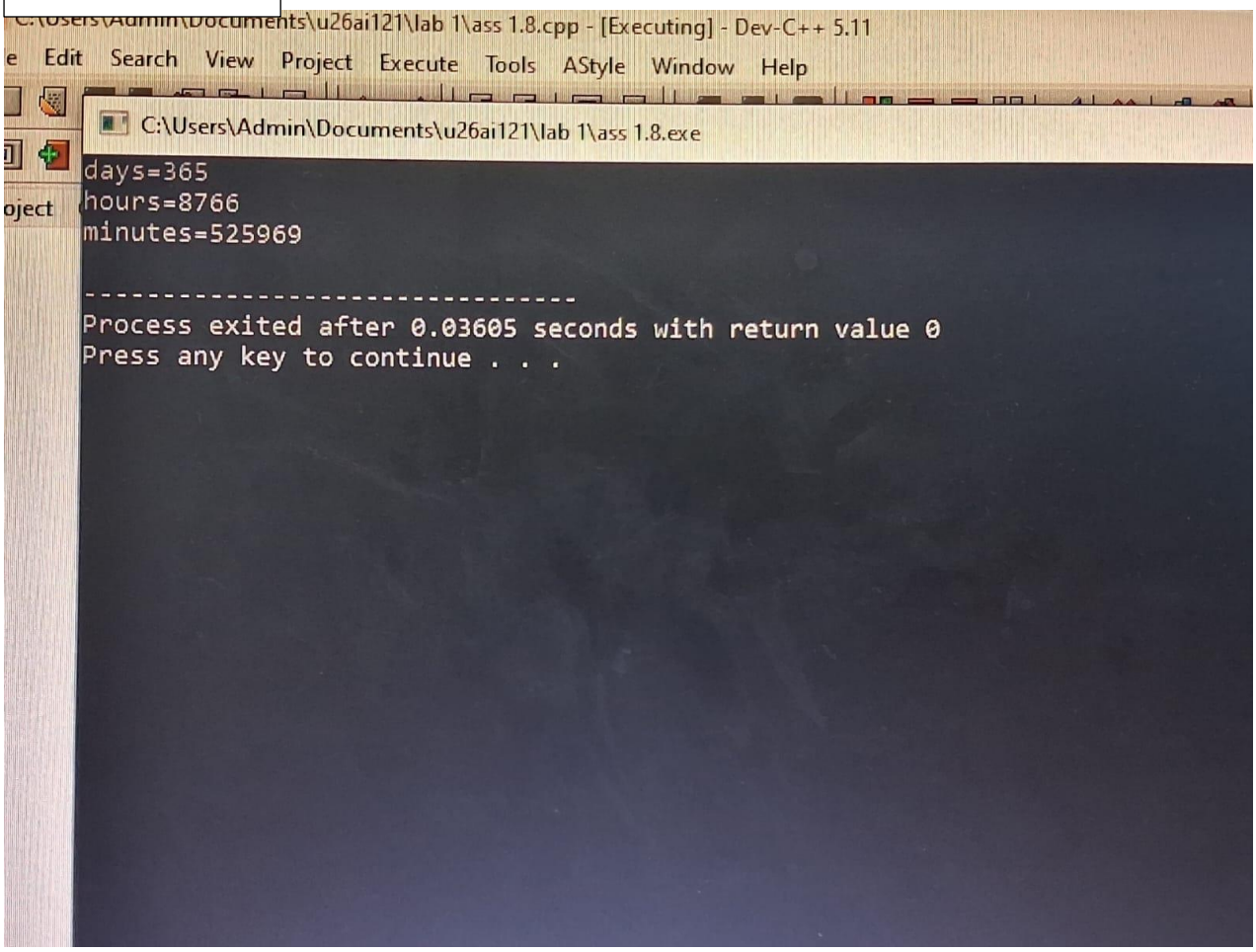


Q.7

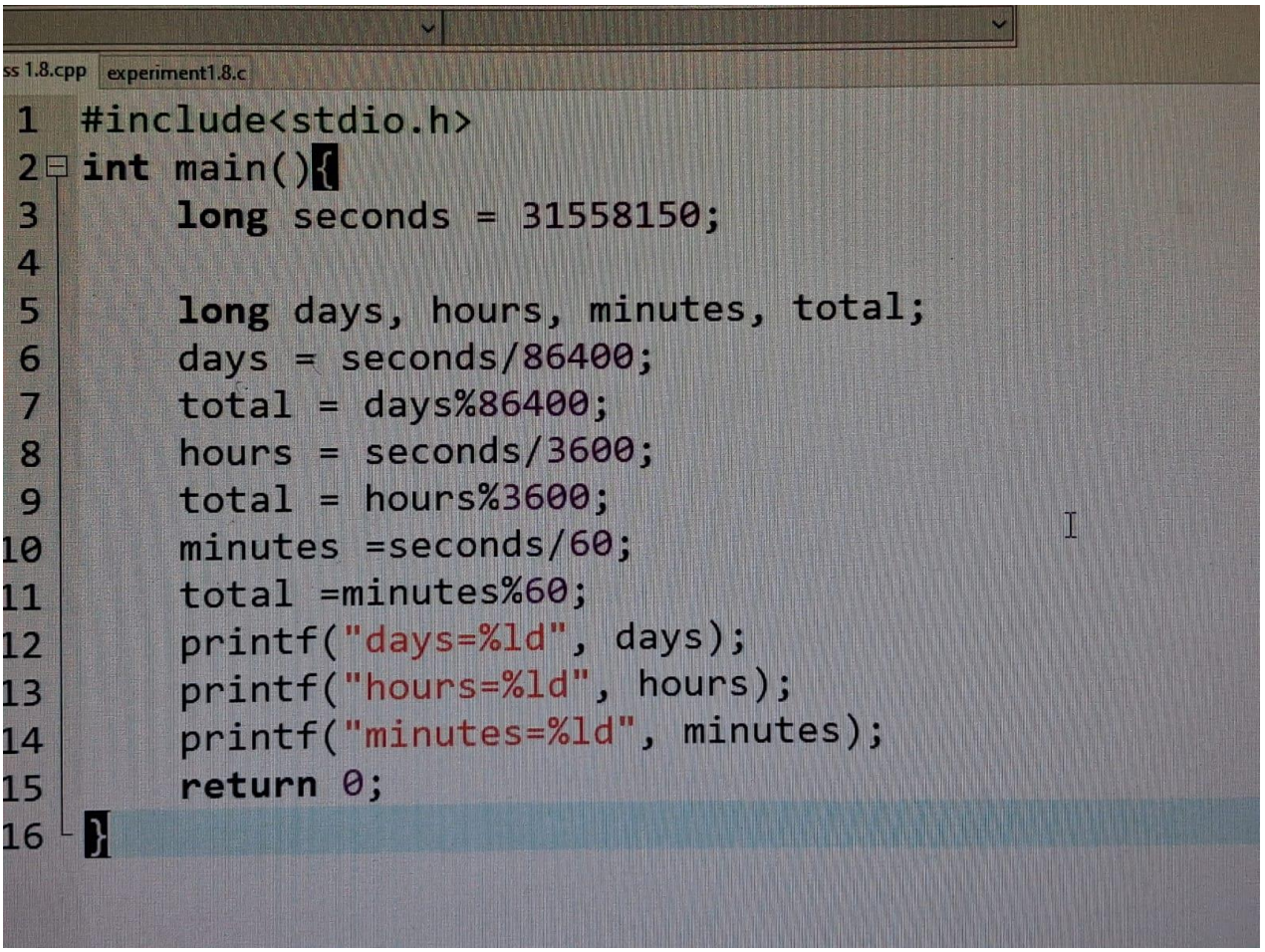



```c
#include <stdio.h>
int main()
{
    float a,b;
    printf("what is the height of triangle:");
    scanf("%f",&a);
    printf("enter the value of triangle's base: ");
    scanf("%f",&b);
   float  c=(a*b)/2;
    printf(" the area of triangle is:%f",c);
    return 0;
}
```

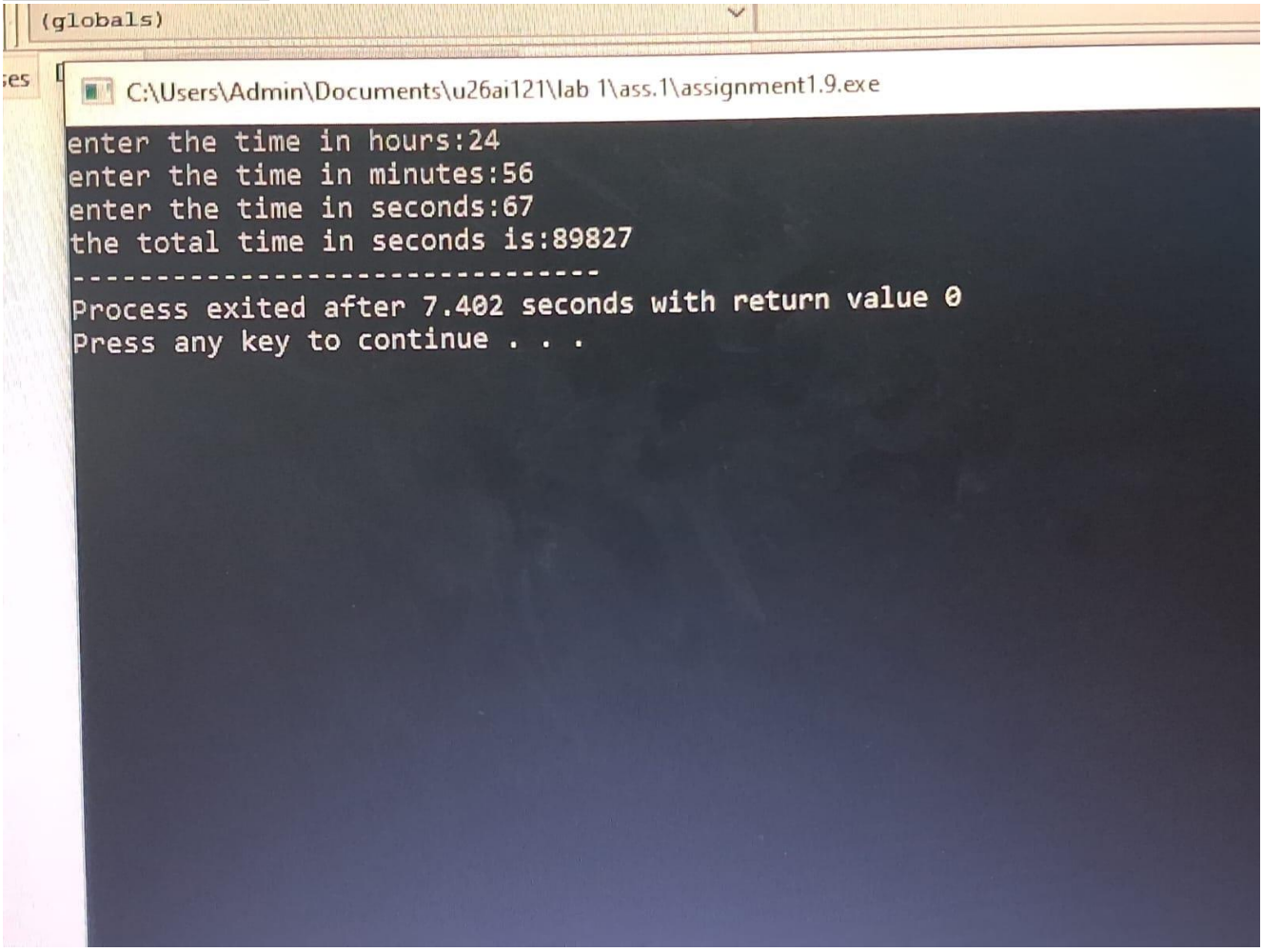
Q.8


C:\Users\Admin\Documents\u26ai121\lab 1\ass 1.8.cpp - [Executing] - Dev-C++ 5.11
e Edit Search View Project Execute Tools AStyle Window Help
C:\Users\Admin\Documents\u26ai121\lab 1\ass 1.8.exe
days=365
hours=8766
minutes=525969

Process exited after 0.03605 seconds with return value 0
Press any key to continue . . .


ss 1.8.cpp
experiment1.8.c
#include<stdio.h>
int main(){
long seconds = 31558150;
long days, hours, minutes, total;
days = seconds/86400;
total = days%86400;
hours = seconds/3600;
total = hours%3600;
minutes =seconds/60;
total =minutes%60;
printf("days=%ld", days);
printf("hours=%ld", hours);
printf("minutes=%ld", minutes);
return 0;
}

Q.9




```c
#include <stdio.h>

int main() {
    int hour,minute,second;
    printf("enter the time in hours:");
    scanf("%d",&hour);
    printf("enter the time in minutes:");
    scanf("%d",&minute);
    printf("enter the time in seconds:");
    scanf("%d",&second);
    int sec=(hour*60*60+minute*60+second);
    printf("the total time in seconds is:%d",sec);


    return 0;
}
```

Q.10

```
\u26ai121\lab 1\ass.1\assignment1.10.cpp - Dev-C++ 5.11
roject  Execute  Tools  AStyle  Window  Help
TDM-GCC 4.9.2 64-bit Release
ass 1.8.cpp | experiment1.8.c | assignment1.9.cpp | assignment1.10.cpp
```

```c
int main() {
    float ch,ma,ph,ex,pr;
    printf("enter your marks in chemistry:\n");
    scanf("%f",&ch);
    if(ch>200 || ch<0 ){
    printf("invalid marks\n");
}

printf("\nenter your marks in math's:\n");
    scanf("%f",&ma);
    if(ma>200 ||ma<0 ){
    printf("invalid marks\n");
}

    printf("\nenter your marks in physics:\n");
    scanf("%f",&ph);
    if(ph>200 || ph<0 ){
    printf("invalid marks\n");
}
    printf("\nenter your marks in entrance examination:\n");
    scanf("%f",&ex);
    if(ex>100 || ex<0 ){
    printf("invalid marks\n");
}
float cm=(ch/2+ma/2+ph/2+ex);
printf("\nthe cutoff marks are:%f\n",cm);
    return 0;
}
```

```
Compile Log  Debug  Find Results
Sel: 0  Lines: 31  Length: 701  Insert  Done parsing in 0.016 seconds
```

```
C:\Users\Admin\Documents\u26ai121\lab 1\ass.1\assignment1.10.exe
enter your marks in chemistry:
78

enter your marks in math's:
78

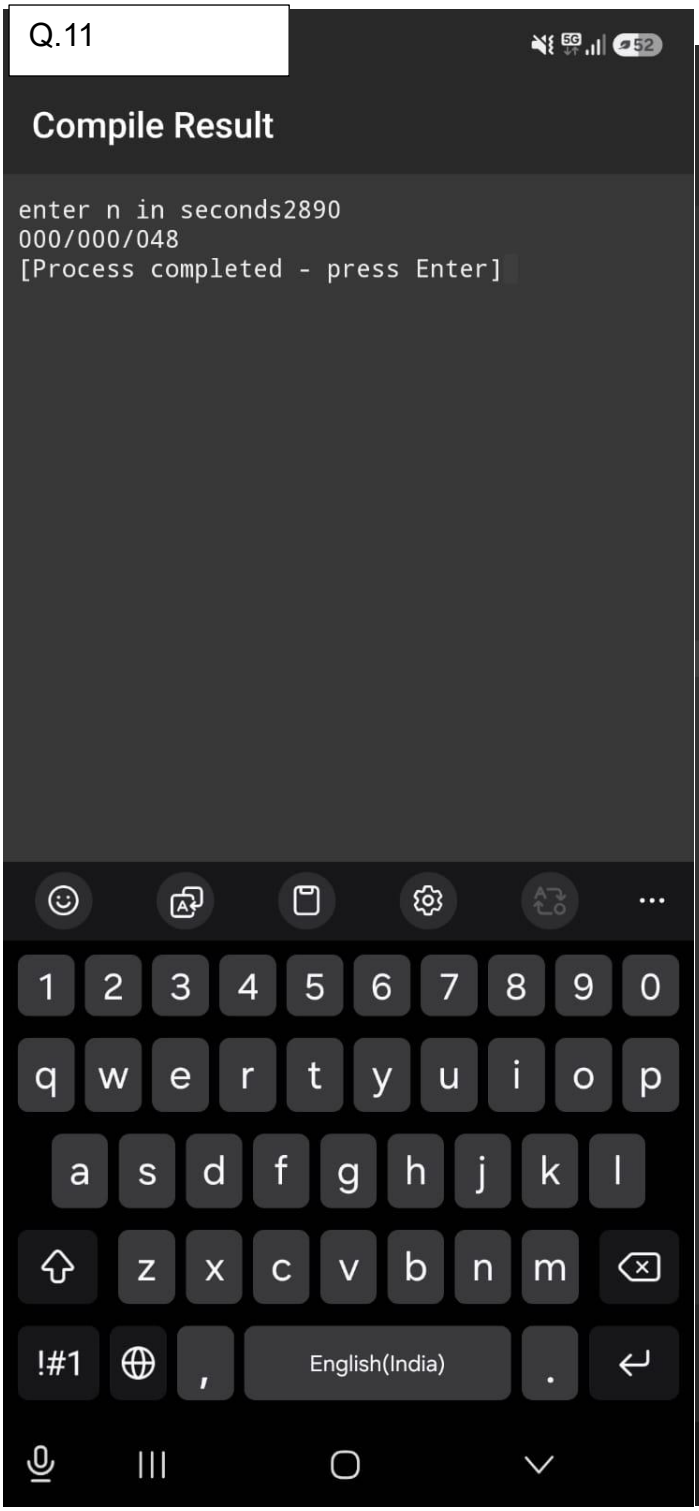
enter your marks in physics:
90

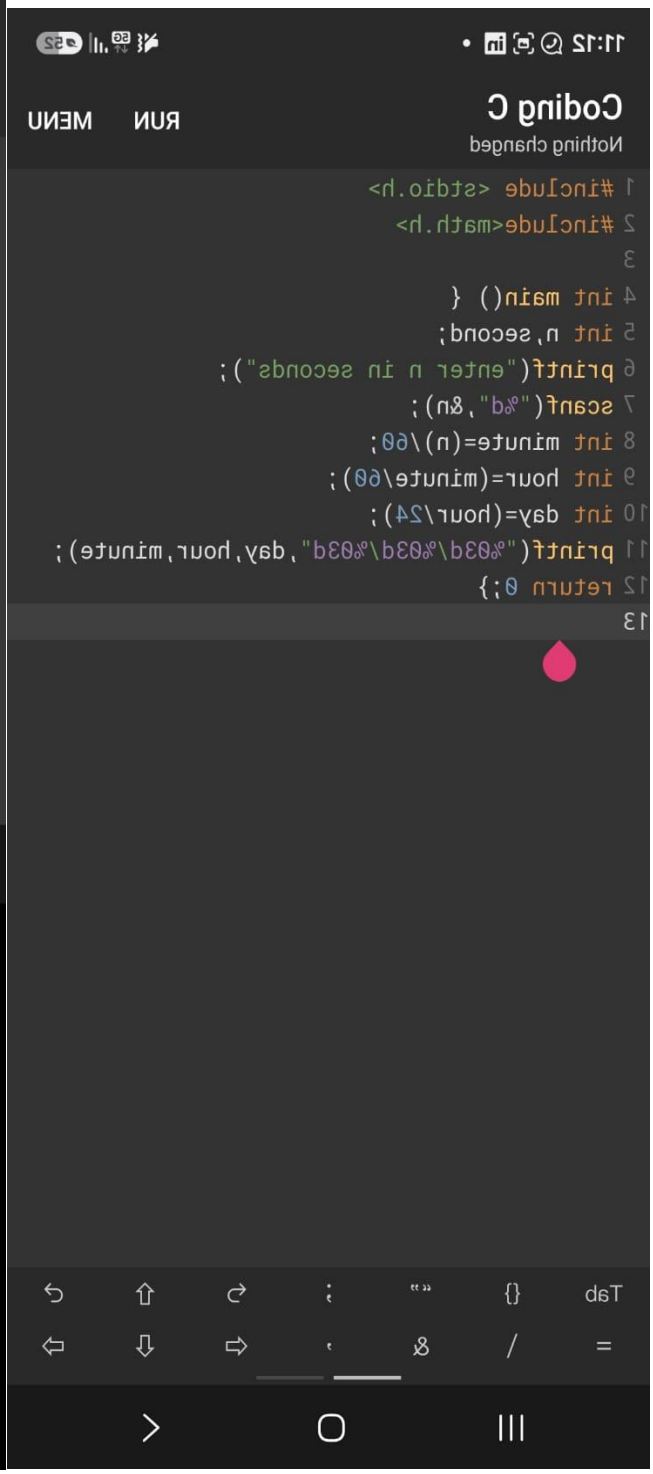
enter your marks in entrance examination:
60

the cutoff marks are:183.000000

--------------------------------
Process exited after 11.02 seconds with return value 0
Press any key to continue . . .
```

Q.11


Compile Result
enter n in seconds2890
000/000/048
[Process completed - press Enter]
English(India)


Coding C
Nothing changed
RUN
MENU
#include <stdio.h>
#include<math.h>
int main() {
int n,second;
printf("enter n in seconds");
scanf("%d",&n);
int minute=(n)\60;
int hour=(minute\60);
int day=(hour\24);
printf("%03d\%03d\%03d",day,hour,minute);
return 0;}

Q.12

```
1
2  #include<stdio.h>
3  int main()
4
5  {char n;
6  printf("enter the n");
7  scanf("%c",&n);
8  printf("the ascii value of %c is %d",n,n);
9   return 0;
0  }
1
```

```
enter the nA
the ascii value of A is 65
--------------------------------
Process exited after 5.669 seconds with return value 0
Press any key to continue . . .
```

Q.13

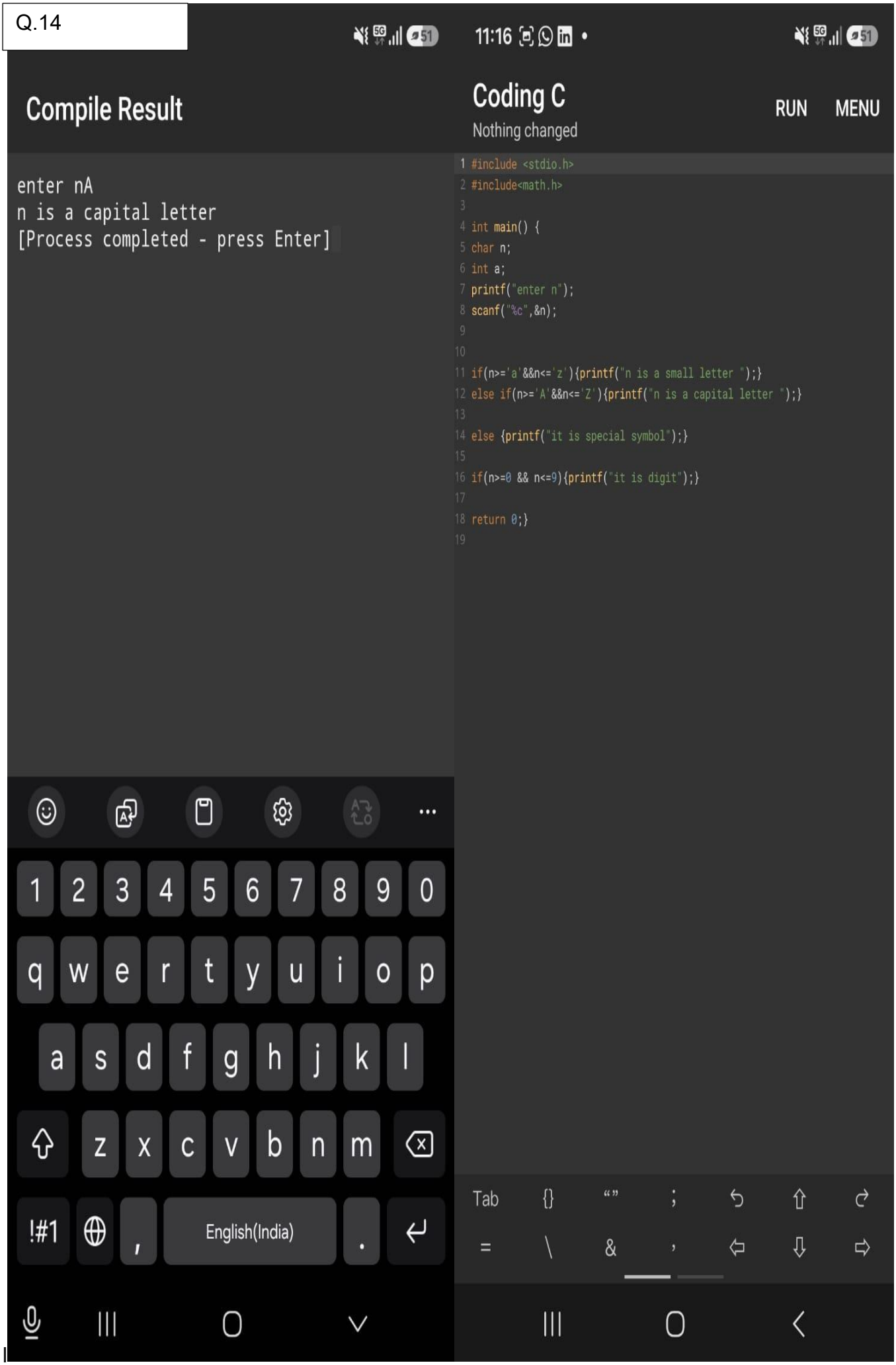
1.11.cpp [*] Untitled1 assignment1.13.cpp assignment1.14.cpp

C:\Users\Admin\Documents\u26ai121\lab 1\ass.1\assignment1.11.exe

```
enter the number n:67
number is odd
--------------------------------
Process exited after 2.223 seconds with return value 0
Press any key to continue . . .
```

```c
#include<stdio.h>
int main()
{int n;
printf("enter the number n:");
scanf("%d",&n);
if(n%2==0){printf("number is even");
}
else if(n%2!=0){printf("number is odd");
}
else{printf("number is zero");
}
 return 0;
}
```

Q.14


Compile Result
enter nA
n is a capital letter
[Process completed - press Enter]
11:16
Coding C
Nothing changed
RUN
MENU
#include <stdio.h>
#include<math.h>
int main() {
char n;
int a;
printf("enter n");
scanf("%c",&n);
if(n>='a'&&n<='z'){printf("n is a small letter ");}
else if(n>='A'&&n<='Z'){printf("n is a capital letter ");}
else {printf("it is special symbol");}
if(n>=0 && n<=9){printf("it is digit");}
return 0;}
English(India)
Tab

Q.15

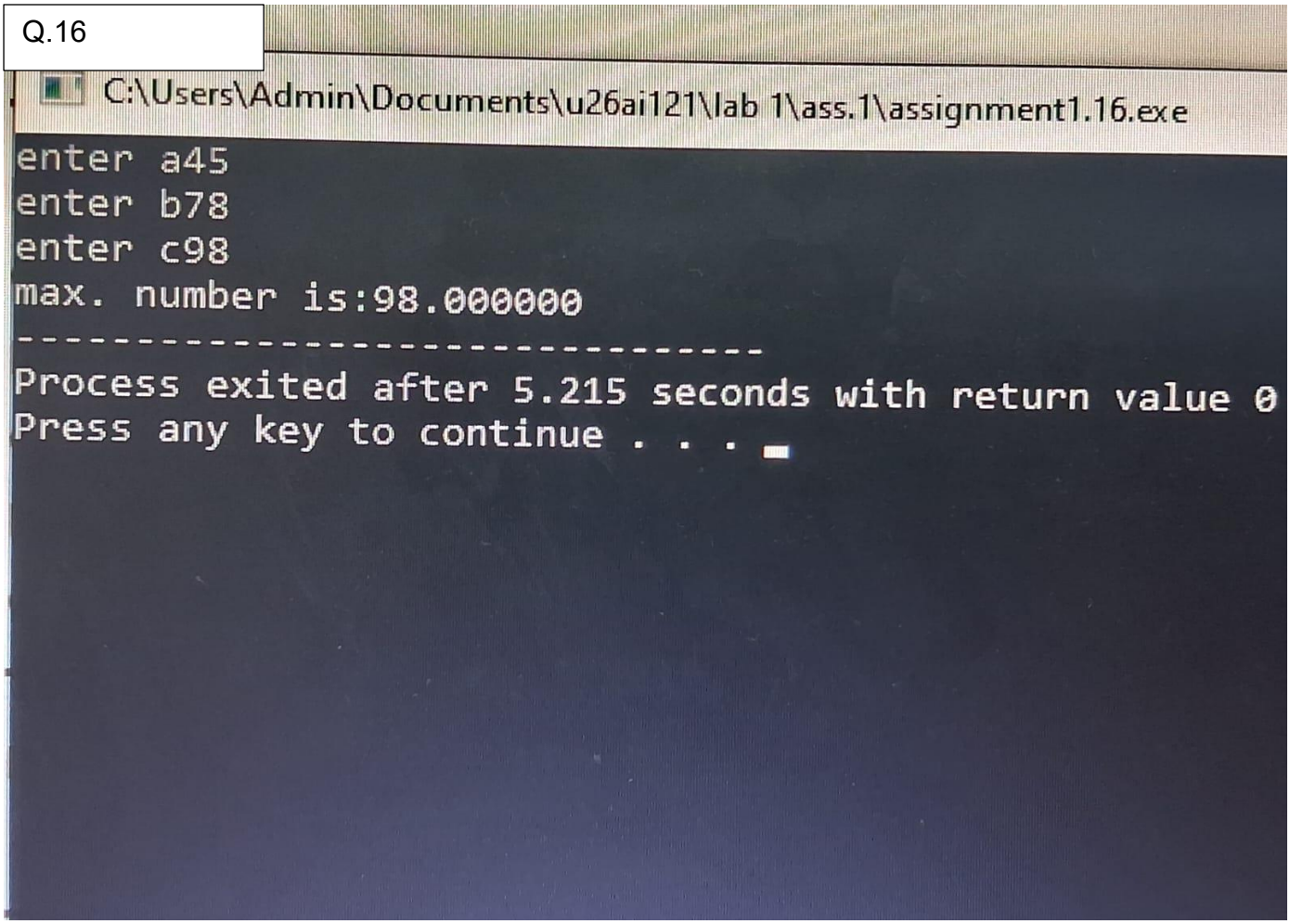
```
...hin\Documents\u26ai121\lab 1\ass.1\assignment1.15.exe
enter marks of subject1:67
enter marks of subject2:78
enter marks of subject3:78
enter marks of subject4:98
enter marks of subject5:69
division a
--------------------------------
Process exited after 9.164 seconds with return value 0
Press any key to continue . . .
```

```c
#include<stdio.h>
int main()

{float subject1,subject2,subject3,subject4,subject5;
printf("enter marks of subject1:");
scanf("%f",&subject1);
printf("enter marks of subject2:");
scanf("%f",&subject2);
printf("enter marks of subject3:");
scanf("%f",&subject3);
printf("enter marks of subject4:");
scanf("%f",&subject4);
printf("enter marks of subject5:");
scanf("%f",&subject5);
float percantage=(subject1+subject2+subject3+subject4+subject5)/5;
if(100>=percantage && percantage>=70){printf("division a");
}
else if(70>percantage && percantage>=50){printf("division b");
}
else if(50>percantage && percantage>=33){printf("division c");
}
else if(33>percantage && percantage>=0){printf("fail");
}
else{printf("not valid");}

 return 0;
}
```

Q.16




```c
#include<stdio.h>
int main()

{float a,b,c;
 printf("enter a");
 scanf("%f",&a);
 printf("enter b");
 scanf("%f",&b);
 printf("enter c");
 scanf("%f",&c);
if(a>b && a>c){
 printf("max. number is:%f",a);
 }
if(b>c){
 printf("max. number is:%f",b);
 }
else{
 printf("max. number is:%f",c);
 }
  return 0;
 }

```

Q.17

```
#include<stdio.h>
int main()

{float a,b,c;
printf("enter a");
scanf("%f",&a);
printf("enter b");
scanf("%f",&b);
printf("enter c");
scanf("%f",&c);
((a>b) && (a>c))?printf("a is the greTEST NUMBER"):printf("");
((b>c) && (b>a))?printf("b is the gretest number"):printf("");
((c>b) && (c>a))?printf("c is the gretest number"):printf("");

 return 0;
}
```

```
int main()
{
p
s
p
s
p
s
(
(
(
```

```
C:\Users\Admin\Documents\u26ai121\lab 1\ass.1\assignment1.17.exe
enter a78
enter b89
enter c90
c is the gretest number
--------------------------------
Process exited after 4.428 seconds with return value 0
Press any key to continue . . .
```

Q.18

```
s\Admin\Documents\u26ai121\lab 1\ass.1\assignment1.18.exe
enter a89
enter b768
enter c68
b is the gretest number
--------------------------------
Process exited after 5.676 seconds with return value 0
Press any key to continue . . .
```

TDM-GCC 4.9.2 64-bit Release

assignment1.11.cpp | [*] Untitled1 | assignment1.13.cpp | assignment1.14.cpp | assignment1.15.cpp | assignment1.16.cpp | assignment1.17.cpp | assignment1.18.cpp | assignment1.

```
1
2  #include<stdio.h>
3  int main()
4
5  {float a,b,c;
6  printf("enter a");
7  scanf("%f",&a);
8  printf("enter b");
9  scanf("%f",&b);
10 printf("enter c");
11 scanf("%f",&c);
12 ((a>b)&&(a>c))?printf("a is the greATEST NUMBER"):printf("");
13 ((b>c) && (b>a))?printf("b is the gretest number"):printf("");
14 ((c>b) && (c>a))?printf("c is the gretest number"):printf("");
15
16   return 0;
17 }
18
```

Q.19

Coding C

Nothing changed

RUN MENU

```c
#include<stdio.h>
int main(){
char alphabet,a,z,A,Z;
printf("enter character alphabet");
scanf("%c",&alphabet);
if(alphabet>='a' && alphabet<='z'){printf("it is small letter");}
else if(alphabet>='A' && alphabet<='Z'){printf("it is capital letter");}
else{printf("it is not letter");}
return 0;}
```

Compile Result

```
enter character alphabetA
it is capital letter
[Process completed - press Enter]
```

Q.20

Compile Result

```
enter a
98
enter b
67
now which arithmetic operations do you want
to do
*
6566

[Process completed - press Enter]
```

Coding C

Nothing changed

RUN MENU

```c
#include<stdio.h>
int main(){
int a,b;
char cal;
printf("enter a\n");
scanf("%d",&a);
printf("enter b\n");
scanf("%d",&b);
printf("now which arithmetic operations do you want to do\n");
scanf("\n%c",&cal);
switch(cal)
{
case '+' :printf("%d\n",a+b);
break;
case '-' :printf("%d\n",a-b);
break;
case '*' :printf("%d\n",a*b);
break;
case '/' :printf("%d\n",a/b);
break;
case '%' :printf("%d\n",a%b);
break;
default :printf("not available");

}
return 0;}
```

Q.21

Coding C

Nothing changed

RUN MENU

```c
#include<stdio.h>
int main(){
int a,b;
char cal;
printf("enter a\n");
scanf("%d",&a);
printf("enter b\n");
scanf("%d",&b);
printf("now which arithmetic operations do you want to do\n");
scanf("\n%c",&cal);
switch(cal)
{
case '+' :printf("%d\n",a+b);
break;
case '-' :printf("%d\n",a-b);
break;
case '*' :printf("%d\n",a*b);
break;
case '/' :printf("%d\n",a/b);
break;
case '%' :printf("%d\n",a%b);
break;
default :printf("not available");

}
return 0;}
```

Tab {} “” ; = \ & ,

5:29

Compile Result

```
enter a
34
enter b
56
now which arithmetic operations do you want
to do
-
-22

[Process completed - press Enter]
```

Q.22

Coding C

Nothing changed

RUN MENU

```c
#include<stdio.h>
int main(){
int n,factorial=1;

printf("enter n");
scanf("%d",&n);
for( int i=1;i<=n;i++){
factorial=factorial*i;
}
printf("%d",factorial);

return 0;}
```

Compile Result

```
enter n7
5040
[Process completed - press Enter]
```

Q.23

Compile Result

```
enter p:5
enter the power that you want:3
answer is 125
[Process completed - press Enter]
```

Coding C

Nothing changed

RUN MENU

```c
#include<stdio.h>
int main(){
int p,n,answer=1;
printf("enter p:");
scanf("%d",&p);
printf("enter the power that you want:");
scanf("%d",&n);
for(int i=1;i<=n;i++){
answer=answer*p;
}
printf("answer is %d",answer);

return 0;}
```

Tab {} "" ; = \ & ,

Q.24

Compile Result

```
enter n3.2
y=1+3.200000x
[Process completed - press Enter]
```

5:31

Coding C

Nothing changed

RUN MENU

```c
#include<stdio.h>
#include<math.h>
int main(){
float n;
char x;
printf("enter n");
scanf("%f",&n);
if(n==1){printf("y=1+x");}
else if(n==2){
printf("y=1+x/%f",n);}
else if(n==3){ printf("y=1+x³");}
else{printf("y=1+%fx",n);}

return 0;}
```

Q.25

Compile Result

```
enter n5
5×1=5
5×2=10
5×3=15
5×4=20
5×5=25
5×6=30
5×7=35
5×8=40
5×9=45
5×10=50

[Process completed - press Enter]
```

5:32

Coding C

Nothing changed

RUN MENU

```c
#include<stdio.h>
#include<math.h>
int main(){
int n;
printf("enter n");
scanf("%d",&n);
for(int i=1;i<=10;i++)
{
printf("%d×%d=%d\n",n,i,n*i);
}

return 0;}
```

Tab {} “” ; = \ & ,

Q.26

Coding C

Auto saved at 17:32:41

RUN MENU

```c
#include<stdio.h>
#include<math.h>
int main(){
int n,s1=0,s2=0;
printf("enter n");
scanf("%d",&n);
for(int i=1;i<=n;i++)
{
if(i%2==0){
 s1=s1+i;}
else{
 s2=s2+i;}
}
printf("sum of even number is=%d\n",s1);
printf("sum of odd number is=%d\n",s2);
return 0;}
```

Tab {} "" ; = \ & ,

5:32

Compile Result

```
enter n50
sum of even number is=650
sum of odd number is=625

[Process completed - press Enter]
```

1 2 3 4 5 6 7 8 9 0

q w e r t y u i o p

a s d f g h j k l

z x c v b n m

!#1 English(India)